

First Principles

Why does S3 have 11 9s of durability but only 4 9s of availability?

Durability = will my data survive?

Availability = can I read it right now?

AWS replicates across 3+ AZs so your data is safe from hardware failure. But any S3 service issue means temporary inaccessibility. Your data is safe but might be briefly unreachable. These are two different engineering problems.

Why is versioning a prerequisite for Cross-Region Replication?

CRR needs to track which versions have been replicated. Without version IDs, S3 can't tell if an object was updated or is the same as what's already copied.

What problem does object storage (S3) solve that block storage (EBS) doesn't?

S3 scales to unlimited size without provisioning. EBS you choose a size upfront. S3 is accessed via HTTP (any client, anywhere). EBS requires mounting to an OS. S3 trades latency for scale, you can't run a database on it.

Why does S3-IA charge retrieval fees but Standard doesn't?

AWS prices reflect access patterns. IA storage is cheaper because AWS optimizes physical placement for infrequent access. When you access it unexpectedly, it costs AWS more to serve and they pass that cost on.

Why is EBS limited to one EC2 instance?

Block storage needs a file system on top. File systems assume exclusive access to blocks.

Two writers to the same blocks without a cluster-aware file system = data corruption. This is a fundamental constraint of block storage, not just an AWS limitation.

Why did AWS create gp3 when gp2 existed?

gp2 tied IOPS to volume size (3 IOPS per GB). Want more IOPS? Buy more storage you don't need. gp3 decouples IOPS from size — pay for what you use. This is cloud economics: decouple resources so customers only pay for what they need.

If GFS is slower and more expensive than EBS, why does it exist?

Some workloads require shared access whether via web server content across an ASG, shared config files, or ML training data. The alternative is syncing files manually or using S3 (not a POSIX filesystem). EFS solves a different problem than EBS.

Why doesn't AWS just make EBS mountable to multiple instances?

Block storage consistency. NFS (what EFS uses) has a protocol for handling concurrent access and file locking. Raw block devices don't. Two writers to the same blocks without coordination = corruption.

Why use separate buckets instead of prefixes with different lifecycle rules?

Separate buckets = separate billing, separate IAM policies, separate encryption keys, clearer governance. In enterprise / defense, bucket boundaries are security boundaries.

Why not just use Intelligent-Tiering for everything?

Monitoring fees add up at scale. For known patterns (ie. archives are never accessed), explicit lifecycle rules are cheaper. I-T is for genuinely unpredictable patterns.